

Password Strength Validator — Project Overview

Hello everyone,

This project is an evolving command-line password validation tool designed to analyze user-submitted passwords for security, structure, and overall strength. The program evaluates passwords based on industry-standard criteria, including length, character diversity, and complexity requirements. It provides clear feedback, strength ratings, and user-friendly messaging to help users create stronger, more secure passwords.

The long-term goal of this project is to grow into a fully featured password utility that includes:

- **Real-time password validation**
- **Strength scoring and recommendations**
- **Retry loops and user-friendly error handling**
- **Color-coded terminal output**
- **A menu-based interface for multiple features**
- **A secure password generator**
- **Optional hidden password input (like real login systems)**
- **Logging and reporting features**
- **Modularized code structure for scalability**

This project is intentionally versioned and will evolve over time, with each update adding new functionality, improved structure, and more professional polish. It serves both as a learning tool and as a portfolio piece demonstrating growth in Python, CLI design, logic structuring, and software development practices. I will also be providing a clean code file alongside a code file with personal notes to show my thought process and how I keep track of things as I am coding. Also if you want I can also add my note pages on what I learned on each version so it's clear what the differences are when adding the new features and will make it clear on my growth. Come along the journey as I continue to grow my skills.